

Cyber Security Lab Exercise

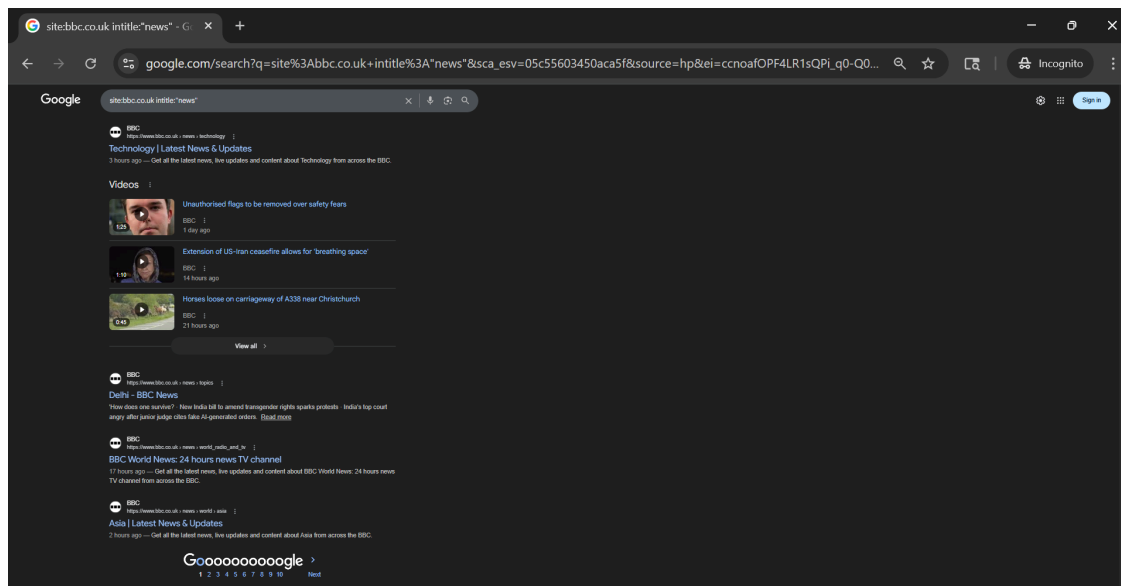
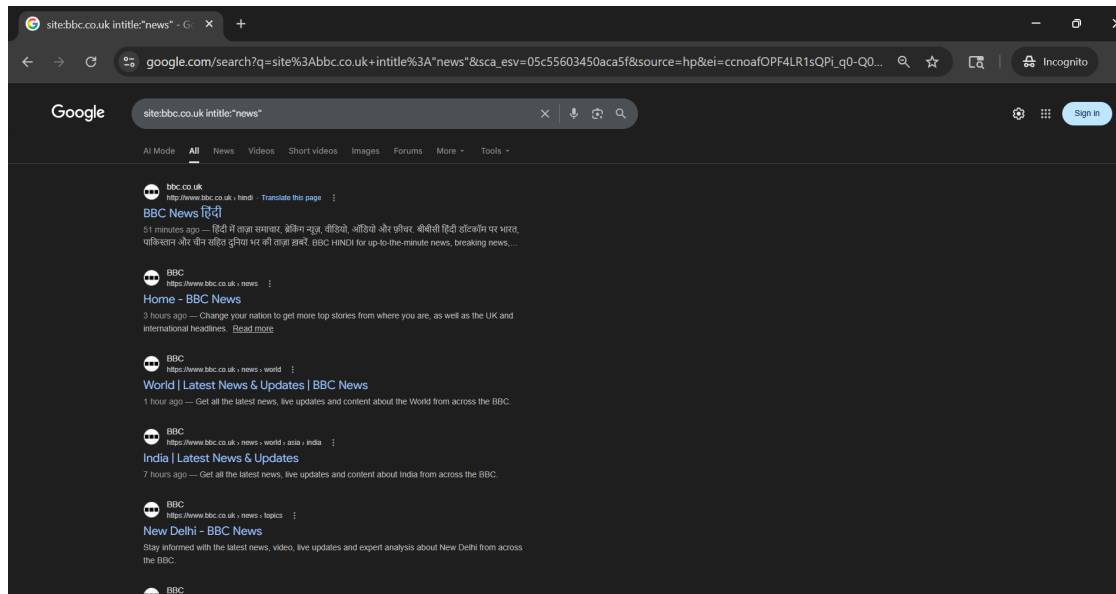
Reconnaissance: Google Dorking

Name: Harshit Kumar More

Roll No: 16

Task 1: Basic Search Operator Practice

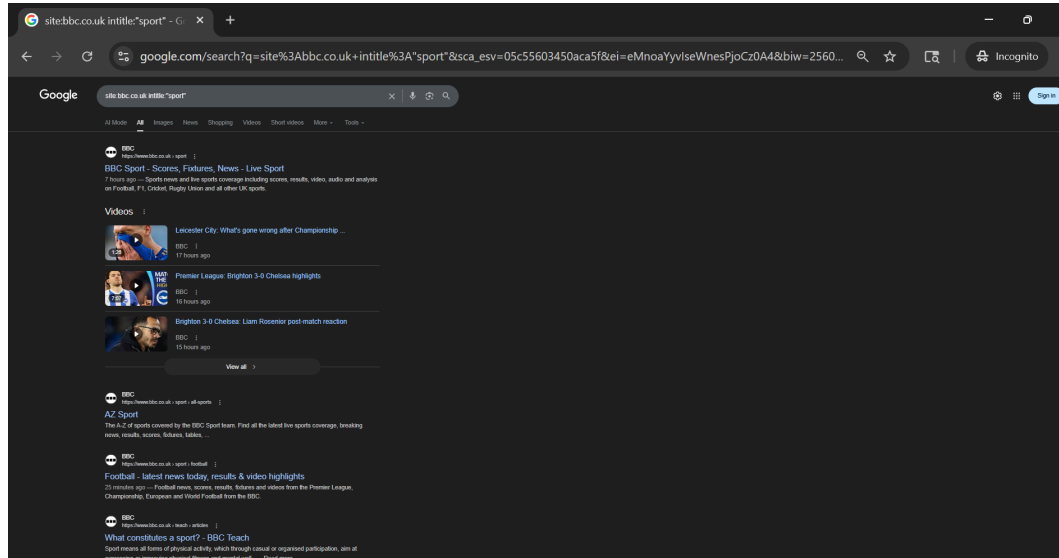
site:bbc.co.uk intitle:"news"



Observation :

The results were restricted to the domain bbc.co.uk. All returned pages contained “news” in their titles. The query behaved as expected, filtering results based on both domain and title.

Modify Query : site:bbc.co.uk intitle:"sport"



After modifying the query to “sport”, the results changed accordingly. The pages now focused on sports-related content, confirming correct query behavior.

7. What is the difference between intitle: and inurl:? Give an example of when you would use each.

The `intitle:` operator searches for keywords in the webpage title, while `inurl:` searches for keywords within the URL.

Example: `intitle:"login"` finds login pages by title, whereas `inurl:admin` finds URLs containing “admin”.

8. Why does using these operators not require any special tools or access?

These operators work directly in Google Search, which indexes publicly available data. Therefore, no additional tools or access are required.

9. How many results did Google return? What does a high result count suggest about that site's content?

Google returned approximately 1,66,00,000 results. A high result count suggests that the website contains a large amount of indexed content.



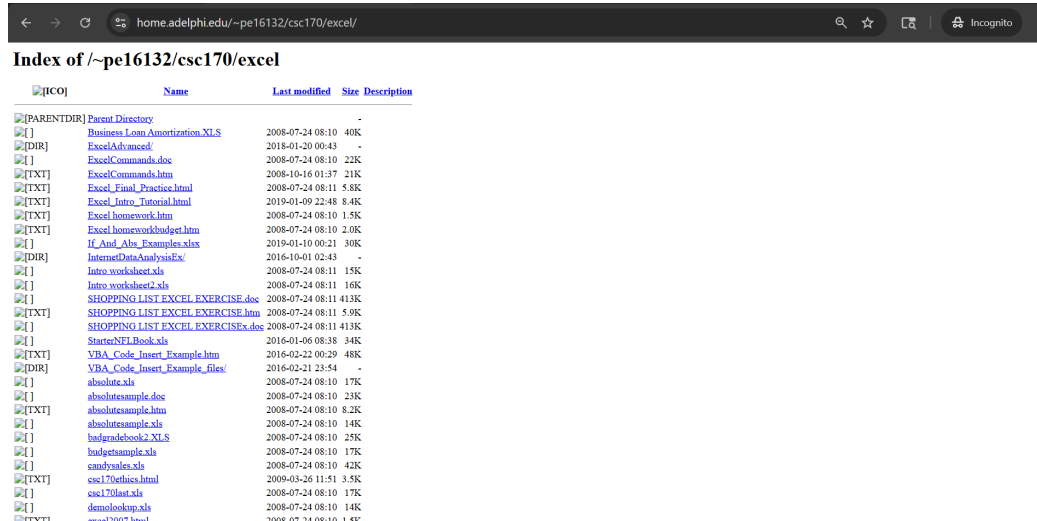
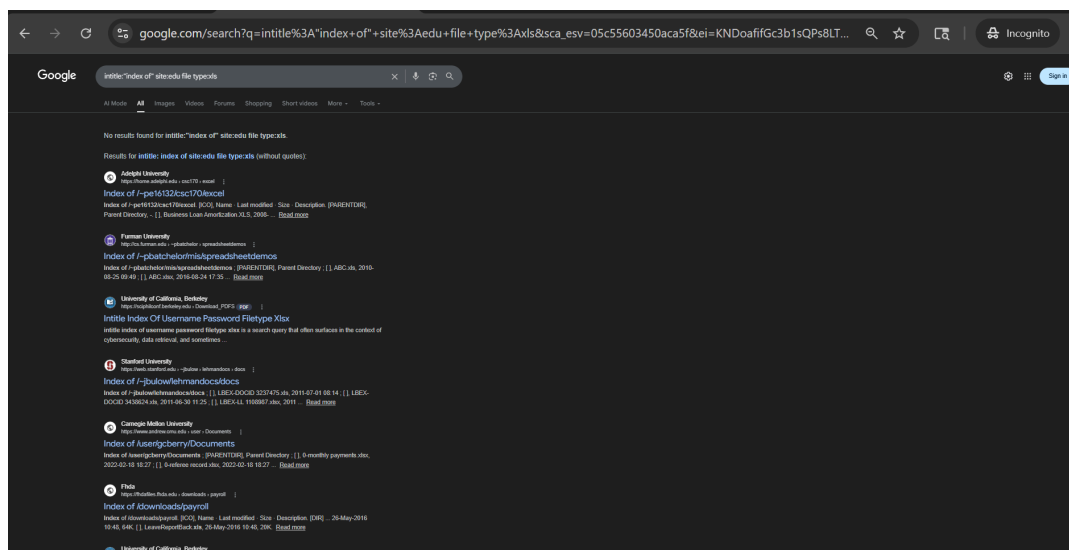
10. Modify the query to search for 'sport' instead of 'news'. Did the results change as expected? Record what you observed.

Yes, the results changed as expected. The new results contained sports-related pages instead of general news.

Task 2: Finding Exposed Files & Systems

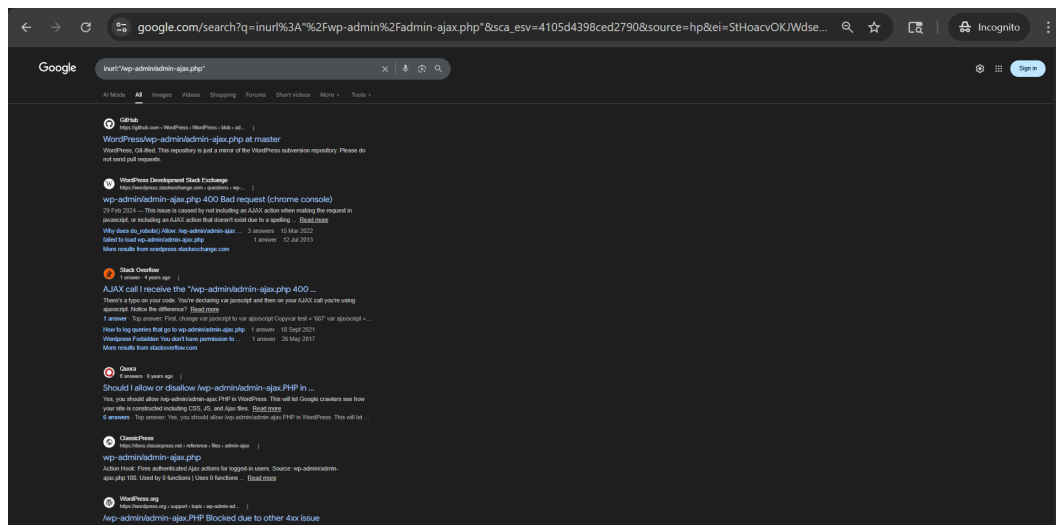
Step 1: Find Exposed Directory Listings on Educational Sites

intitle:"index of" site:edu filetype:xls



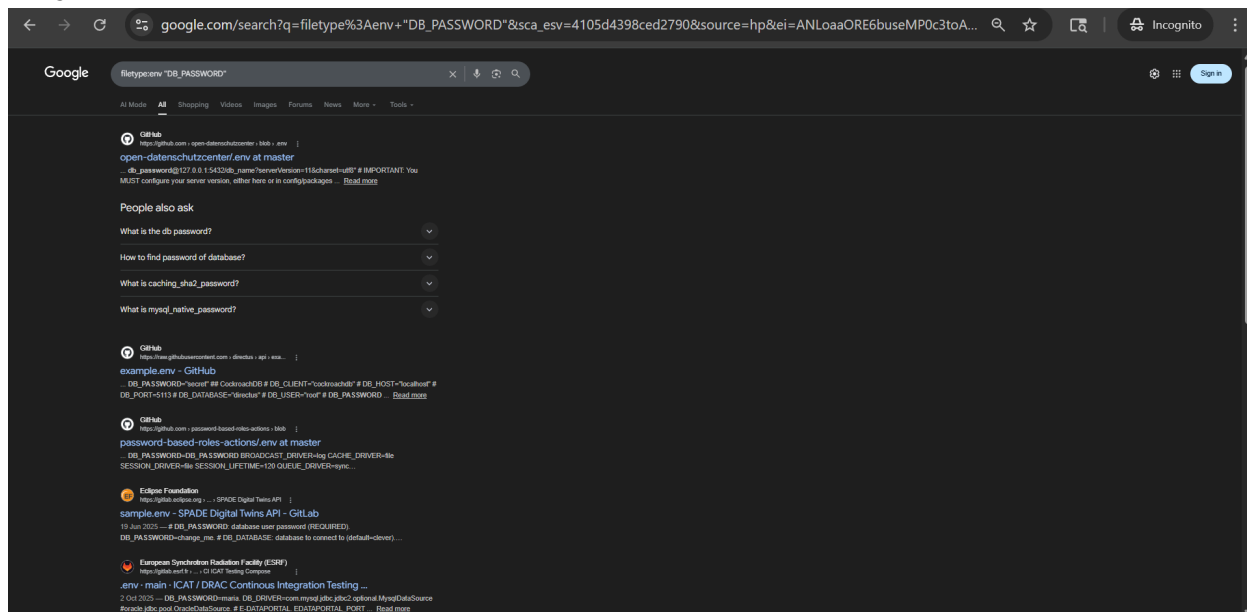
The results showed directory listings from educational websites containing Excel files. File names suggested possible sensitive data such as student records and budgets.

Step 2: Find Exposed WordPress Admin Panels inurl:"/wp-admin/admin-ajax.php"



The results displayed contain URLs containing WordPress admin paths, indicating publicly accessible backend endpoints

Step 3: Find Exposed Environment Configuration Files filetype:env "DB_PASSWORD"

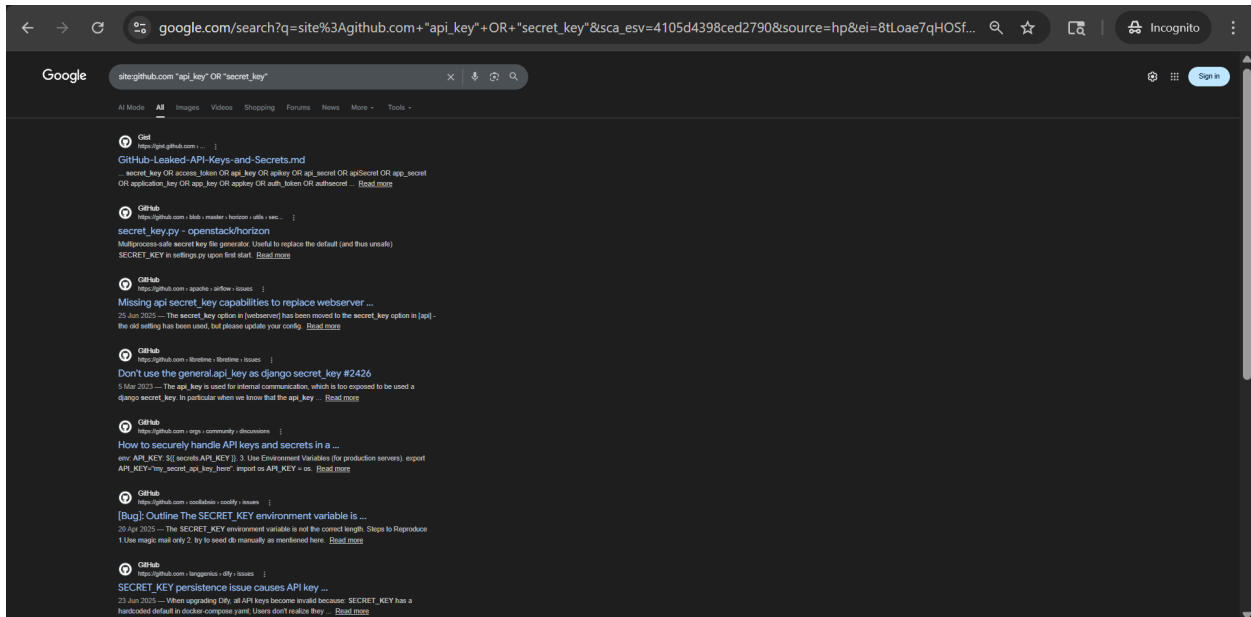


```
raw.githubusercontent.com/directus/directus/f26d862a35c587cd3fd6f15d8e9b6469d16489c7/api/example.env

#####
# General
HOST="0.0.0.0"
PORT=8095
PUBLIC_URL="http://localhost:8095"
LOG_LEVEL="info"
LOG_STYLE="pretty"
#####
# Database
## These match the databases defined in the docker-compose file in the root of this repo
## Postgres
DB_CLIENT="Pg"
DB_HOST="localhost"
DB_PORT=5438
DB_DATABASE="directus"
DB_USER="postgres"
DB_PASSWORD="secret"
## CockroachDB
DB_CLIENT="cockroachdb"
DB_HOST="localhost"
DB_PORT=5113
DB_DATABASE="directus"
DB_USER="root"
DB_PASSWORD=""
## MySQL
DB_CLIENT="mysql"
DB_HOST="localhost"
DB_PORT=5181
DB_DATABASE="directus"
DB_USER="root"
DB_PASSWORD="secret"
## MariaDB
DB_CLIENT="mariadb"
DB_HOST="localhost"
DB_PORT=5182
DB_DATABASE="directus"
DB_USER="root"
DB_PASSWORD="secret"
## MS SQL
DB_CLIENT="mssql"
DB_HOST="localhost"
DB_PORT=5183
DB_DATABASE="directus"
DB_USER="sa"
DB_PASSWORD="Test@123"
## OracleDB
DB_CLIENT="oracledb"
DB_CONNECT_STRING="localhost:1518/XE"
#####
```

Some results indicated exposed environment configuration files. Snippets showed variable names such as DB_PASSWORD, suggesting potential credential exposure.

Step 4: Find Leaked API Keys on GitHub
site:github.com "api_key" OR "secret_key"



Results included code repositories containing references to API keys or secret keys. Some appeared to be real, while others were placeholders.

Step 5: Document Your Findings

Dork Query	Exposure Type	Risk Level	Screenshot Taken?
intitle:"index of" site:edu filetype:xls	Open Directory Listing	High	Yes
inurl:"/wp-admin/admin-ajax.php"	Admin Panel Exposure	Medium	Yes
filetype:env "DB_PASSWORD"	Sensitive File Exposure	Critical	Yes
site:github.com "api_key" OR "secret_key"	Credential Leakage	Critical	Yes

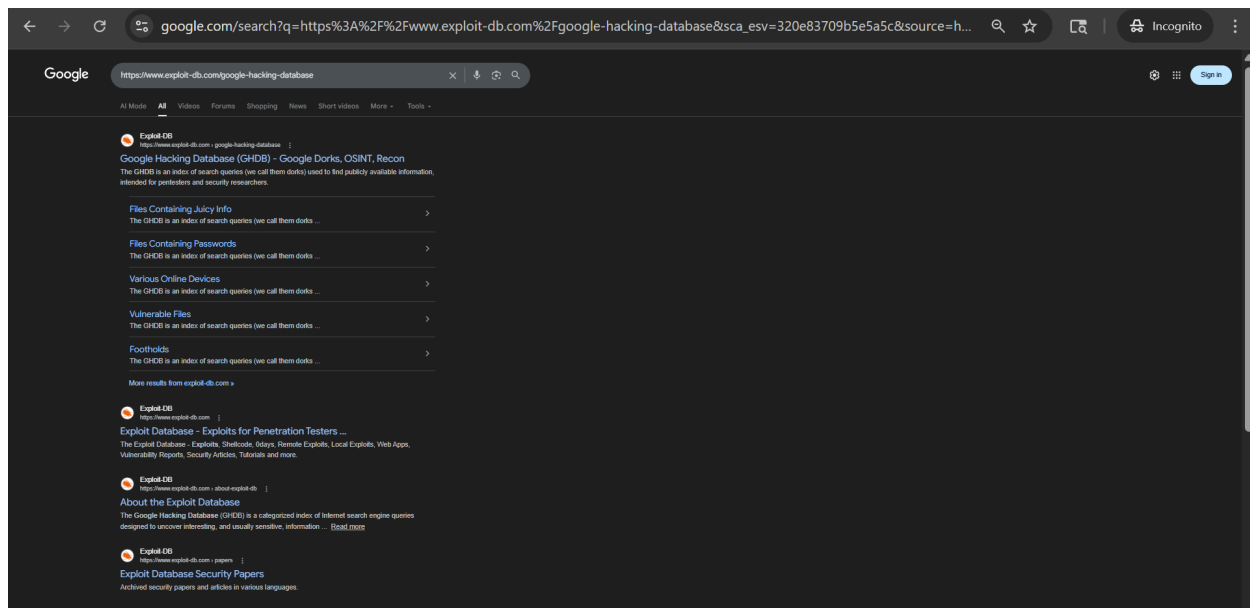
11. Which of the four queries returned the most results? Why do you think that type of exposure is so widespread?

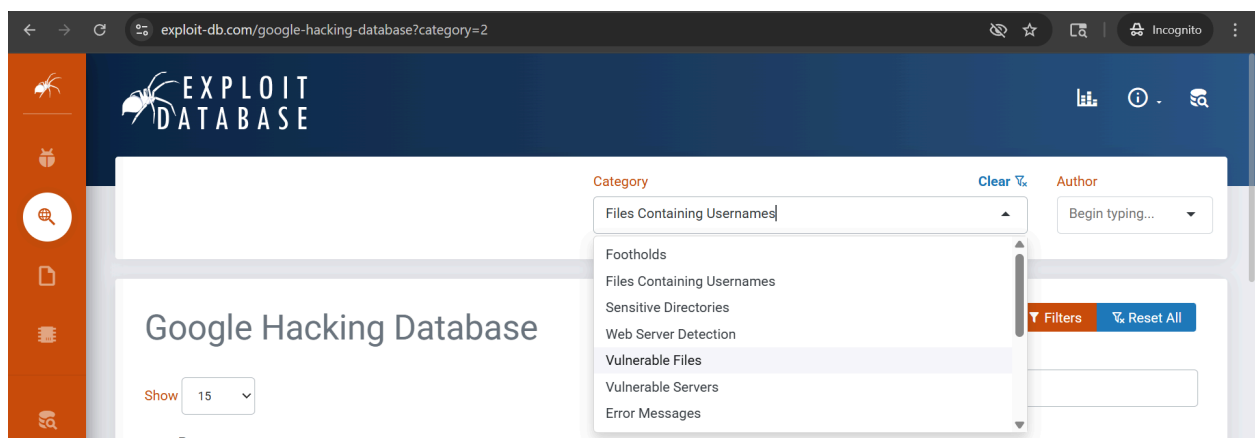
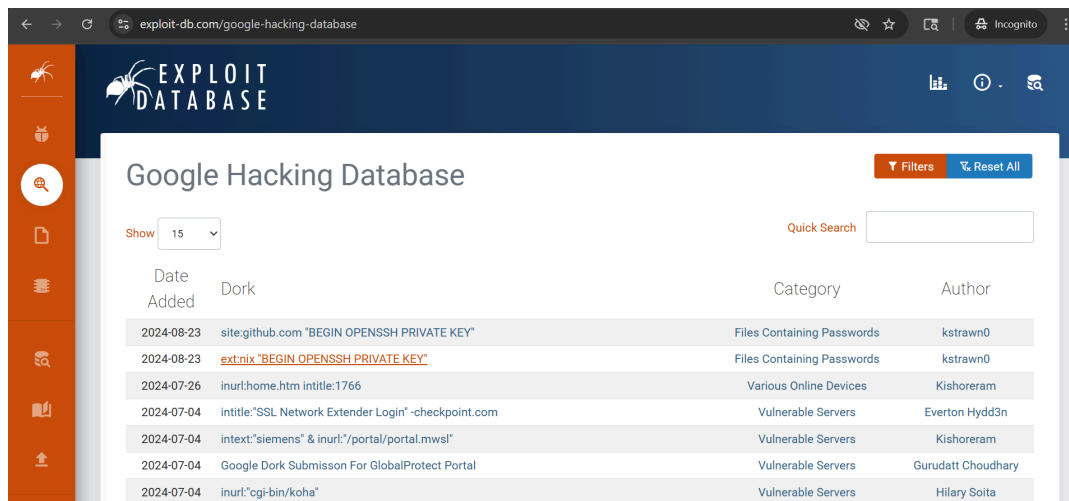
The GitHub API key query returned the most results. This is because developers frequently upload code publicly and sometimes include sensitive information unintentionally.

12. What does the presence of exposed .env files tell you about developer security awareness in general?

The presence of exposed .env files indicates poor security awareness among developers. It shows a lack of proper configuration management and secure coding practices.

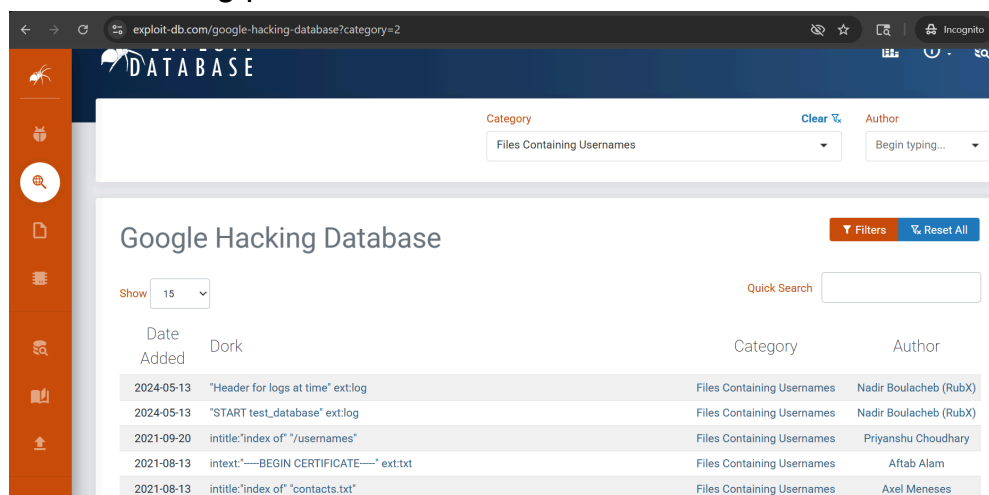
Task 3: Google Hacking Database (GHDB)



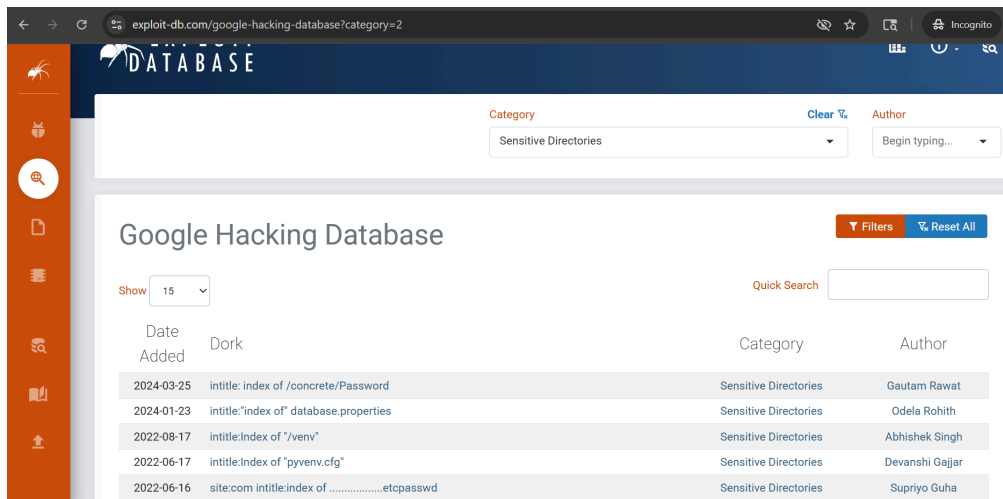


Read at least three entries per category and observe how operators are combined:

1) File Containing password:



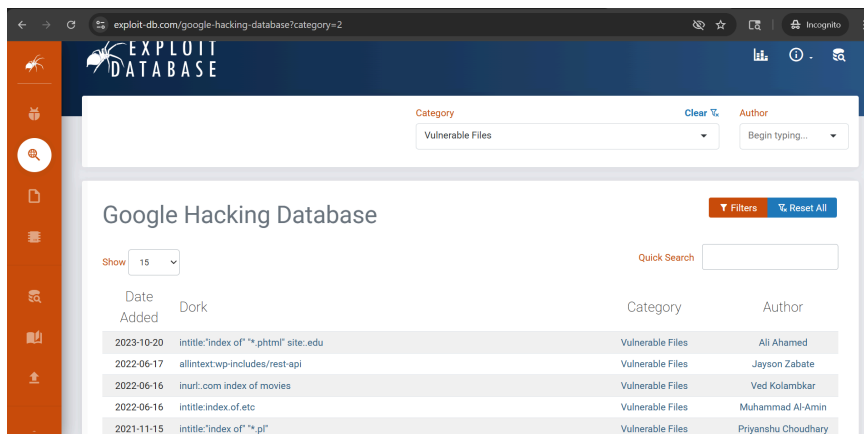
2) Sensitive Directories



The screenshot shows the Exploit-DB Google Hacking Database interface. The 'Category' dropdown is set to 'Sensitive Directories'. The 'Author' dropdown is set to 'Begin typing...'. The 'Show' dropdown is set to '15'. The 'Quick Search' input field is empty. The table displays the following data:

Date Added	Dork	Category	Author
2024-03-25	intitle:index of /concrete/Password	Sensitive Directories	Gautam Rawat
2024-01-23	intitle:"index of" database.properties	Sensitive Directories	Odela Rohith
2022-08-17	intitle:index of "/.env"	Sensitive Directories	Abhishek Singh
2022-06-17	intitle:index of "pyenv.cfg"	Sensitive Directories	Devanshi Gajjar
2022-06-16	site:com intitle:index ofetcpasswd	Sensitive Directories	Supriyo Guha

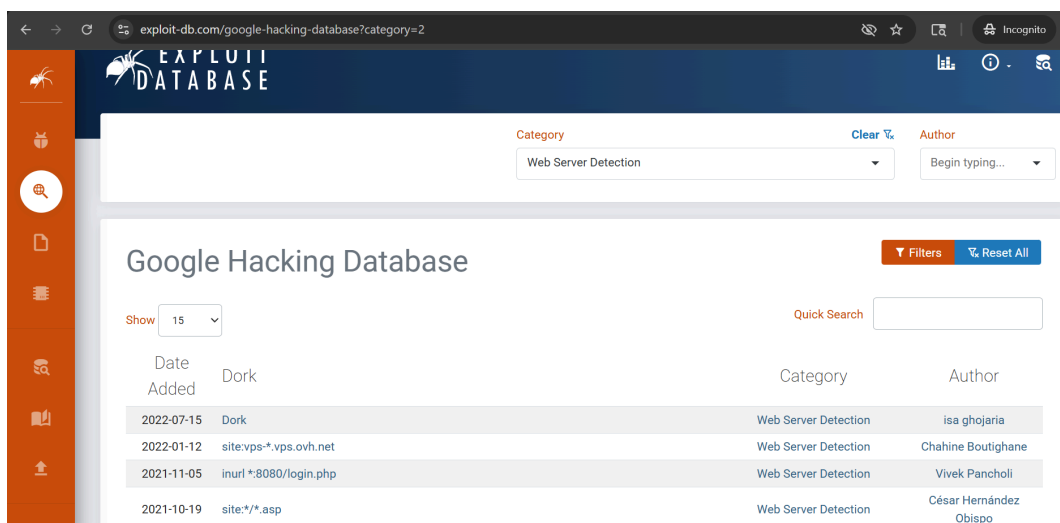
3) Vulnerable Files



The screenshot shows the Exploit-DB Google Hacking Database interface. The 'Category' dropdown is set to 'Vulnerable Files'. The 'Author' dropdown is set to 'Begin typing...'. The 'Show' dropdown is set to '15'. The 'Quick Search' input field is empty. The table displays the following data:

Date Added	Dork	Category	Author
2023-10-20	intitle:"index of" *.phtml" site:.edu	Vulnerable Files	Ali Ahamed
2022-06-17	allintext:wp-includes/rest-api	Vulnerable Files	Jayson Zabate
2022-06-16	inurl:.com index of movies	Vulnerable Files	Ved Kolambkar
2022-06-16	intitle:index of .etc	Vulnerable Files	Muhammad Al-Amin
2021-11-15	intitle:"index of" *.pl"	Vulnerable Files	Priyanshu Choudhary

4) Web Server Detection



The screenshot shows the Exploit-DB Google Hacking Database interface. The 'Category' dropdown is set to 'Web Server Detection'. The 'Author' dropdown is set to 'Begin typing...'. The 'Show' dropdown is set to '15'. The 'Quick Search' input field is empty. The table displays the following data:

Date Added	Dork	Category	Author
2022-07-15	Dork	Web Server Detection	isa ghajarja
2022-01-12	site:vps-*.vps.ovh.net	Web Server Detection	Chahine Boutighane
2021-11-05	inurl *.8080/login.php	Web Server Detection	Vivek Pancholi
2021-10-19	site:*/*.asp	Web Server Detection	César Hernández Obispo

Step 3: Run Three Dorks of Your Own Choice

Category	Dork Query Used	What Was Found	Risk Level
Vulnerable File	<code>inurl:.com index of movies</code>	Results showed directory listings containing folders or files with names related to passwords. This suggests possible exposure of sensitive credentials in server directories.	Critical
Web Server Detection	<code>intitle:"index of" site:.gov.in</code>	Results displayed open directory listings on government domains. These directories revealed file structures and sometimes documents accessible without authentication.	High
Sensitive Directories	<code>intitle: index of /concrete/Password</code>	Results showed publicly accessible directories containing media files such as movies. This indicates improper server configuration allowing file browsing.	Medium

← → ↻ google.com/search?q=%09intitle%3A"index+of"+site%3A.gov.in&sca_esv=320e83709b5e5a5c&biw=2560&bih=1170&ei=6OXoadb9... 🔍 ☆ 📄 Incognito ⋮

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Index of Images
Index of Images : [PARENTDIR] Parent Directory [IMG] 6 x 3 - banner_teachers_day.jpg, 2025-10-29 17:56 [IMG] 1080x slow.gif, 2025-10-29 17:56 ... [Read more](#)

**DCME LP**
[http://lpme.up.gov.in/assets/admin](#)

Index of assets/admin
Index of assets/admin: [PARENTDIR] Parent Directory [IMG] 4025x165x150.jpg, 2021-02-11 23:07 [IMG] Suresh Kumar Sharma.jpg, 2021-12-22 06:15 ... [Read more](#)

**NERPC**
[http://nerpc.gov.in/wp-includes/wp-content/Proxy](#)

Index of /wp-includes/Requests/sr/Proxy
12 Mar 2025 --- Index of /wp-includes/Requests/sr/Proxy [ICO] Name Last modified Size Description [PARENTDIR] Parent Directory ... [Read more](#)

**NERPC**
[http://nerpc.gov.in/wp-includes/wp-content/admin-ui](#)

Index of /wp-includes/wp-content/admin-ui
Index of /wp-includes/wp-content/admin-ui: [PARENTDIR] Parent Directory [TXT] style-rtl.css, 2025-03-12 17:24 [TXT] style-rtl.min.css, 2025-03-12 17:24 ... [Read more](#)

**Ministry of Statistics and Program Implementation**
[http://msoip.gov.in/wordpress/wp](#)

Index of Industrial Production (IP) recorded a 4.8 % year- ...
25 Jan 2025 --- The Quick Estimate of Index of Industrial Production (IP) is released on 28th of every month (or next working day if 28th is a Holiday) ... [Read more](#)
7 pages

**Centre for Cultural Resources and Training (CCRT)**
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Index of /wp-content/uploads
26 Feb 2025 --- Index of /wp-content/uploads [PARENTDIR] Parent Directory [DIR] 2026/ 2026-04-01 05:30 [DIR] ccb7_uploads/ 2026-02-26 16:22 ... [Read more](#)

**DCME LP**
[http://lpme.up.gov.in/assets/dist/img](#)

Index of assets/dist/img
Index of assets/dist/img: [PARENTDIR] Parent Directory [IMG] aviator.png, 2016-05-13 12:26 [IMG]

← → ↻ google.com/search?q=inurl%3A.com+index+of+movies&sca_esv=320e83709b5e5a5c&biw=2560&bih=1170&ei=NuboaZ_vG5Cese... 🔍 ☆ 📄 Incognito ⋮

Google X | 📄 🔍

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Index of /movies
Index of /movies: Name Last modified Size Description Parent Directory - 3pgipvid.mp4, 2016-07-26 03:26, 2.8M, 345x241x101 12:58 ... 37m.mpg ... [Read more](#)

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[http://play.google.com/store/apps/details?id=com...](#)

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3.2 ★★★★★ (138) Free Android Entertainment

**cinelink.com**
[http://www.cinelink.com/this/index](#)

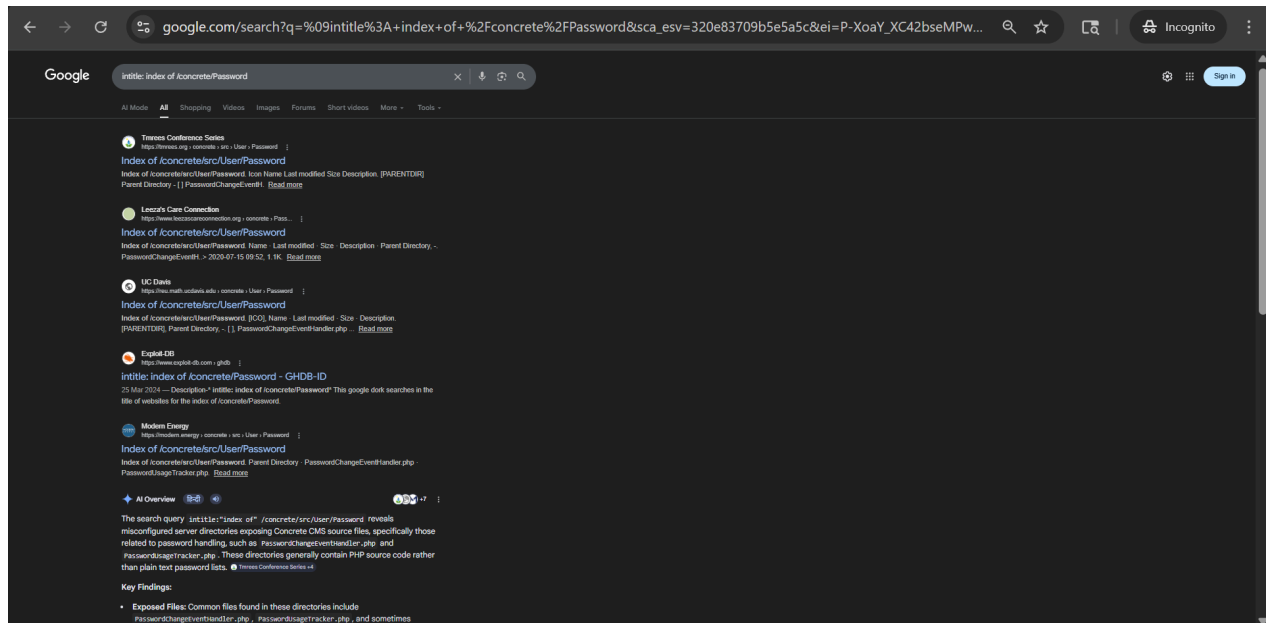
Index of /films/india
Index of /films/india [VID] Lab-Tak-Hai-Jaan2012.mp4, 2025-03-17 08:18 [VID] Mankanda's The Queen of Jhansi2019.mp4, 2025-03-17 08:34 [1] RoseKingdom ... [Read more](#)

**Lemon TV**
[http://lch.bencix.com/movies](#)

Index of /movies
Index of /movies [ICO] Name Last modified Size Description [PARENTDIR] Parent Directory - [DIR] GDI+ New, 2019-07-11 09:46 - [DIR] Dealer ... [Read more](#)

**Stanford University**
[http://stanford.edu/movies](#)

Index of /movies
Index of /movies [ICO] Name Last modified Size Description [DIR] Parent Directory - [VID]



13. Which GHDB category produced the most results when you ran your dork? Why do you think that category has so many entries?

The “Sensitive Directories” category produced the most results. This is because many web servers are misconfigured to allow directory listing, making folder contents publicly visible and easily indexed by search engines.

14. Which category seems most likely to generate false positives, and why?

The “Sensitive Directories” category can produce false positives because not all open directories contain sensitive information. Some may host publicly intended content such as media files.

15. What was the most surprising exposure you found? Explain why it represents a security risk.

The most surprising exposure was open directories on government domains. This is a security risk because it may expose official documents and internal data without proper access controls.

Task 4: Classification, Risk, and Defences

A. Open Directory Listing

Dork used:

intitle:"index of" site:edu filetype:xls

intitle:"index of" site:.gov.in

inurl:.com index of movies

B. Exposed Administrative Surface

Dork used:

inurl:"/wp-admin/admin-ajax.php"

C. Leaked Credentials

Dork used:

```
site:github.com "api_key" OR "secret_key"  
intitle:index of /concrete/Password
```

D. Sensitive File Exposure

Dork used:

```
filetype:env "DB_PASSWORD"
```

Step 2: Evaluate Defensive Countermeasures

16. A developer argues that adding a file to robots.txt is sufficient to stop it being found via Google dorking. Why is this incorrect?

The robots.txt file only instructs search engines not to index certain pages, but it does not provide any actual security. Anyone with the direct URL can still access the resource. Therefore, relying on robots.txt does not prevent attackers from discovering or accessing sensitive information.

17. Why is rotating an exposed credential not enough on its own? What else must the organisation investigate?

Rotating an exposed credential is not sufficient because the credential may have already been used by an attacker. The organization must also review system logs to detect unauthorized access, investigate potential breaches, and identify whether other credentials or systems have been compromised.

18. Explain the difference between deindexing a page and actually securing the underlying resource.

Deindexing removes a page from search engine results, making it less visible to users. However, the resource may still be accessible via its direct URL. Securing a resource involves implementing proper access controls such as authentication, authorization, or firewall restrictions to prevent unauthorized access.

Security Analysis Questions

Q1. A company discovers its internal spreadsheets are appearing in Google dork results. What three immediate steps should the security team take, and what long-term controls should they put in place?

The three immediate steps the security team should take are:

- 1) Remove the exposed files from public access
- 2) Restrict permissions and secure the server
- 3) Rotate any exposed credentials

Long-term controls include implementing proper access control mechanisms, conducting regular security audits, and providing security awareness training to employees to prevent similar incidents.

Q2. How does passive OSINT using Google dorking differ from active reconnaissance techniques such as port scanning? Why is the distinction legally important?

Passive reconnaissance, such as Google dorking, involves gathering information from publicly available sources without directly interacting with the target system. In contrast, active reconnaissance, such as port scanning, involves directly probing systems, which can be detected and may be illegal without authorization. This distinction is legally important because passive methods are generally allowed, while active methods may violate laws or policies

Q3. Why is it insufficient to simply delete an exposed file from a public server after it has already been indexed by Google? What additional steps must be taken?

Simply deleting an exposed file is not sufficient because search engines may have cached copies of the file, and it may still be accessible through those caches or other sources. Additional steps must include requesting removal from search engine indexes, clearing cached versions, and monitoring logs to ensure the file has not been accessed or exploited.